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Imperatives, Interrogatives and Quantifiers in Korean

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Jae-Il Yeom. 2020. Imperatives, interrogatives and quantifiers in Korean. *Language and Information* 24.2, 19-50. In this paper, I discuss restrictions on quantifiers used in interrogatives and imperatives in Korean. Krifka (2001) tried to explain them by assuming that a quantifier quantifies into speech acts and that speech acts are conjoined but not disjoined. I show that speech acts are disjoined under the condition that disjoined speech acts are re-interpreted as a single speech act. In interrogatives, quantifiers cannot have wide scope over a *wh*-phrase. The restriction is that an interrogative with a quantifier must have the meaning of one question. In imperatives, some quantifiers are not allowed because they generate some conventional implicatures which are not compatible with the semantics of imperatives. Other quantifiers are not allowed in imperativs because they are used to express genericity, which is also not compatible with the semantics of imperatives. To show this, I briefly discuss the semantics of imperatives. (**Hongik University**)

Key words: imperative, interrogative, disjunction, conventional implicatures, speech act, genericity

1. Introduction

It was claimed by Krifka (2001) that in English a quantifier can quantify into speech acts. As a supporting example, he gave a case where a question with a quantifier has a pair-list reading. A question with a quantifier can have three readings. Each of them is answered differently:

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(1) Which dish did every guest make?
  a. (Every guest made) pasta. (narrow scope)
  b. (Every guest made) his favorite dish. (functional reading)
  c. Al (made) the pasta; Bill, the salad; and Carl, the pudding. (pair-list reading)

Among the three readings, he claimed that the pair-list reading is obtained when the quantifier has wide scope over the interrogative Mood. The other two readings are possible with the quantifier having narrower scope than the Mood.

Such pair-list readings obtain when a universal quantifier is involved. No pair-list reading is allowed in a question with a non-universal quantifier:[1)]

(2) Which dish did {most, several, a few, no} guests make?
  a. Pasta.
  b. Their favorite dish.
  c. #Al, the pasta, and Bill, the salad.

Krifka (2001) tried to explain that robust pair-list interpretations arise only with universal quantifiers, on the basis of the restriction that speech acts can be conjoined, but not disjoined. Suppose that there are three guests: {A(l), B(ill), C(arl)}. Then interrogatives with *every* and *most* can be interpreted as follows:

(3) Which dish did every guest make? = For every guest x: Which dish did x make?
  = Which dish did A make, <u>and</u>
    which dish did B make, <u>and</u>

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1) Ginzburg & Sag (2000) do not agree with these observations, but their observations are slightly different from Krifka's, and they cannot refute his claim:

i. A: Most people here have submitted a paper to a journal.
   B: Which journal?
   A: Alexis to Psychic Review, Pat to Post-Modern Letters, …

In the discourse, B's utterance does not have to be spelled out as Which journal have most people submitted a paper to? – rather, the first sentence makes available a discourse referent for the people that have submitted a paper to a journal. B's utterance spells out as: Which journal have these people submitted a paper to? Therefore Ginzburg & Sag's example is not a counter-example to Krifka's claim. Consider the following contrast:

ii. a. Here is a list of 20 African countries. (…). Choose at least 11 of them and write down their capitals.
   b. Here is a list of 20 African countries. (…) *Which capital do most of them have?

which dish did C make?

(4) #Which dish did most guests make? = For most guests x: which dish did x make?
= Which dish did A make and which dish did B make, **or**
which dish did A make and which dish did C make, **or**
which dish did B make and which dish did C make?

In (3), the interrogative with the universal quantifier is equivalent to one conjunction of the interrogatives with each individual in the domain of quantification. In (4), however, the interrogative with *most* can be represented with a disjunction of three conjunctions of the interrogatives with each individual in the domain of quantification.

However, Krifka's claim is controversial. First, it is not clear if the restriction on disjunction applies to speech acts or Mood Phrases (= MPs). When Krifka tried to show the restriction on disjunction of speech acts, he actually conjoins or disjoins two MPs, rather than two speech acts. In Korean, no MPs are conjoined or disjoined:[2)]

(5) a. ??inho-ka ttena-ss-ta kuliko mina-ka ttena-ss-ta.[3)]
Inho-nom leave-pst-dec and Mina-nom leave-pst-dec
'Inho left, and Mina left.'
b. inho-ka ttena-ss-ta. kuliko mina-ka ttena-ss-ta.
Inho-nom leave-pst-dec and Mina-nom leave-pst-dec
'Inho came. And Mina came.'

The conjunctive connective *kuliko* 'and' does not combine two MPs. It connects two separate sentences, with each constituting a separate speech act.

Second, it is not clear whether the restriction applies to the semantics or syntax of sentences. A speech act is an act of uttering a sentence. We assume that a sentence has a structure, but we do not assume that an act of uttering does. This leads to a controversy of whether two speech acts can be disjoined. There are cases where two separate questions can be disjoined, as pointed out by Belnap and Steel (1976):

2) In the paper, I use the following abbreviations: acc(usative case), adn(ominal ending), c(o)mp(lementizer), dec(larative mood), hon(orific marker), imp(er)f(ective aspect), imp(erative mood), int(errogative mood), n(o)m(ina)l(izer), nom(inative case), p(a)st(tense), top(ic marker), etc.

3) One anonymous reviewer doubts that simply removing the period at the end of the first MP makes any difference. A sentential punctuation indicates the end of a full speech act. Without a sentential punctuation, the intonation would not fall down completely but the second MP would be able to continue without a full stop. This is not acceptable.

(6) Have you ever been to Sweden or have you ever been to Germany?

About this example, Krifka claims that it is one question that asks whether the addressee has ever been to Sweden or to Germany, and that it can be answered with *yes* or *no*. However, structurally it is clear that two interrogatives are disjoined, regardless of how the disjunction of the two sentences is interpreted.

In this paper, I am more concerned with the second issue. The restriction on disjunction of speech acts is expected to apply to all speech acts, and it is expected to be universal. Then it is expected to apply to interrogatives and imperatives in Korean. In Korean, however, the observations are not consistent with English data, and even within Korean. In interrogatives, no pair-list interpretation obtains, as in English:

(7) {taypwupwunuy, manhun/twu} salam-i nwukwu-lul manna-ss-nya?
{most, many/two} people-nom who-acc meet-pst-int
'Who did {most, many/two} people meet?'
a. *inho-nun mina-lul manna-ss-ko mino-nun unha-lul manna-ss-ko ⋯
Inho-top Mina-acc meet-pst-and Mino-top Unha-acc meet-pst-and ⋯
'Inho met Mina, and Mino met Unha, and ⋯'
b. Inho.
c. caki ayin. 'Their lovers.'

The quantifiers are disjunctive and the *wh*-phrase *nwukwu* must have wide scope over the quantifiers, or the sentences have a functional reading. If the quantifiers in (7) had wide scope over the *wh*-phrase, the meanings would be represented as in (4), and they are unacceptable.[4)]

As for the universal quantifier, there is controversy about the pair-list reading. (8) is odd with the pair-list reading, but (9) is not:

(8) motun salam-i nwukwu-lul manna-ss-nya?
all people-nom who-acc meet-pst-int
'Who did all people meet?'
(9) A: {motwu, ta-tul} eti ka-ss-nya?
{everybody, all-pl} where go-pst-int
'Where has everybody gone?'
B: inho-nun cip-ey, mino-nun PC.pang-ey ... ka-ss-e.

4) Here I only deal with examples in which the predicate is interpreted as distributive.

Inho-top home-to Mino-top internet.cafe-to go-pst-dec
'Inho went home, and Mino went to an internet cafe.'

Kim (1991) does not accept a pair-list reading, but Suh (1990), Joo (1989), and Choi (2002) do. Hoji (1985) and Hagstrom (1998) observe in Japanese that no pair-list reading obtains. I suppose this is due to some vagueness which is not captured by a question-answer pair as in (9). The discourse in (9) is acceptable, but A does not know that people in the context went to different places. If A did, she would not use the interrogative with a universal quantifier. Therefore a pair-list reading is not really speaker's meaning in the sense that the speaker never intends that reading with the sentence. Therefore no pair-list reading is allowed.

This can be compared with a definite NP:

(10) (ku) twu salam-un nwukwu-lul manna-ss-nya?
the two people-top who-acc meet-pst-int
'Who did the two people meet?'

This sentence can be uttered when the speaker knows each person met a different person. This shows that definite NPs are different from universal quantifiers.

In imperatives, even a disjunctive quantifier is allowed:

(11) {motun, ??taypwupwunuy, manhun, (ceketo) twu, ??kyewu twu} salam-i
{all, most, many, (at.least) two, merely two} people-nom
ttena-la!
leave-imp
'{All, Most, Many, At least two, Just two} people leave!'

(12) {motun, taypwupwunuy, manhun, (ceketo) twu, kyewu twu} salam-i
{all, most, many, (at.least) two, merely two} people-nom
ttena-ss-ta.
leave-pst-dec
'{All, Most, Many, At least two, Merely two} people left.'

The imperative with *motun* can be expressed with the conjunction of the imperatives for all individuals of the addressee, but the imperative with the other quantifiers should be represented with the disjunction of conjunctions of the imperatives for some individuals of the addressee.[5] Still quantifiers like *manhun* 'many' and *ceketo twu* 'at least two' are allowed in imperatives, though quantifiers like *taypwupwunuy* 'most' and *kyewu twu* 'at most two' are not.

This means that there is a restriction on quantifiers that can be used in imperatives, but that it is different from the restriction on quantifiers in interrogatives. In this paper, I will discuss what factors are involved in allowing quantifiers in interrogatives and imperatives.

The paper is organized as follows. In Section 2, I discuss conjunction and disjunction of speech acts, and show that interrogatives and imperatives can be disjoined if the disjunctions can be re-interpreted as one speech act, and that conjoined or disjoined phrases can have various scopes, even over the scope of the Mood, unless the result readings are allowed pragmatically. In Section 3, I discuss restrictions on quantifiers in interrogatives and show that restrictions on pair-list readings can be explained by assuming that a quantifier cannot have scope over the Mood in Korean. In Section 4, I claim that an imperative changes the world by the speaker's decision and the addressee's accepting the decision. The semantics of an imperative imposes a semantic restriction on expressions that can be used in imperatives. A quantifier that has a semantic or pragmatic meaning that is incompatible with the meaning of an imperative is not allowed in imperatives. In Section 5, I conclude the paper.

2. Disjunction, conjunction and speech acts

2.1. Conjunction of speech acts

As pointed out in Section 1, statements are conjoined with *kuliko* 'and'. Questions and commands are also conjoined:

(13) a. inho-ka ttena-ss-nya? kuliko mina-ka ttena-ss-nya?
Inho-nom leave-pst-int and Mina-nom leave-pst-int
'Did Inho leave? And Did Mina leave?'
b. inho-ka ttena-la! kuliko mina-ka nam-ala!
Inho-nom leave-imp and Mina-nom stay-imp
'Inho leave! And Mina stay!'

One condition is that the conjunction operator combines the same type of sentences, as observed in Yeom (2019). It is odd to conjoin different speech acts:

5) In this discussion, I assume that the quantifiers are interpreted as distributive. If the predicate is not distributive, the imperatives are not paraphrased as a conjunction of imperatives for individuals in the addressee.

(14) ??inho-ka ttena-ss-nya? kuliko mina-ka nam-ala!
Inho-nom leave-pst-int and Mina-nom stay-imp
'Did Inho leave? And Mina stay!'

I represent the act of uttering a declarative MP "ϕ." as υ(ϕ.), where υ is an act of uttering. Then the condition of the same type of speech act is represented as follows:

(15) υ(ϕ.) kuliko υ′(ψ.); υ(ϕ?) kuliko υ′(ψ?); υ(ϕ!) kuliko υ′(ψ!)
("ϕ.", "ϕ?", "ϕ!": declarative, interrogative, and imperative MP, respectively)

One reason for this condition seems to be that if two speech acts are conjoined, they form a single speech act, which is expressed by a single sentence with a conjunction structure within. This implies that it is not important what phrases are conjoined in expressing the meaning of a conjunction of two speech acts. Thus, the intended meanings of the sentences in (13) can be expressed by the following sentences with the coordination structure interpreted distributively:

(16) a. inho-ka ttena-ko mina-ka ttena-{ss-nya?, la!}
Inho-nom leave-and Mina-nom leave-{pst-int imp}
{'Is it the case that Inho left and Mina left?', 'Inho leave and Mina leave!'}
b. inho-wa mina-ka ttena-{ss-nya?, la!}
Inho-and Mina-nom leave-{pst-int imp}
{'Did Inho and Mina leave?', 'Inho and Mina leave!'}

The conjunction of two questions might be considered different from one question with a conjunction structure within in the ways the context is structured:

(17)

Conjunction of two questions	One question with a conjunction structure
Inho & Mina left	Inho & Mina left
Inho left & Mina did not leave	It is not the case that Inho & Mina left
Inho did not leave & Mina left	
Neither Inho nor Mina left	

If each question partitions the common ground, the conjunction of two questions partitions the common ground into four groups of possible worlds and a question with a conjunction structure partitions it into two groups of possible worlds.

However, this is just one possible interpretation. If each conjunct includes a *wh*-phrase, the question can be interpreted as multiple questions. Suppose that we have to decide one of the four options depending on who Inho is meeting and where Mina is. In this context, we can ask the following question:

(18) [[inho-nun nwukwu-lul manna]-ko [mina-nun eti-ey iss]]-nya?
Inho-top who-acc meet-and Mina-top where-at be-int
'Who is Inho meeting and where is Mina?'

In (18), the two *wh*-phrases are not expected to move out of the conjunction operator at LF, but the speaker asks two questions. This shows that to get the meaning of conjunction of speech acts, we do not need an actual conjunction of speech acts.[6)]

Regardless of what phrases are conjoined, if a conjunction operator or conjoined NP has wide scope over a *wh*-phrase, we can get the meaning of multiple speech acts. This might be surprising considering the fact that a universal quantifier, which is semantically conjunctive, does not lead to a pair-list reading, as discussed in (8) and (9). This difference can be explained by the fact that the scope of a definite NP, conjoined or not, can be scopeless while the scope of a universal quantifier is determined by its syntactic position.[7)] We need to show that a quantifier does not have scope over the Mood. This issue will be discussed below.

2.2. Disjunction of speech acts

Krifka (2001) explains the restriction on quantifiers in interrogatives and imperatives based on the restriction that speech acts are not disjoined. To see if we can resort to the same explanation, we need to consider disjunction of speech acts in Korean. Before I do this, I will discuss a little more of disjunction of speech acts in English. Here is an example of imperatives:

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6) In interpreting a sentence with a conjunction structure, the conjunction operator can be interpreted differently depending on what phrase is conjoined.

7) A coordinating operator can have wide scope over the Mood of a sentence:

i. a. John met Mary or Sue.
b. Implicatures: It is possible that John met Mary & it is possible that John met Sue.

The implicatures obtain if the disjunction operator has wide scope over the whole sentence

(19) Pick up the ball or pick up the racket.
  a. 'Act to make true: You pick up the ball or you pick up the racket.'
  b. 'Pick up the ball, or pick up the racket, I don't know which.'

The disjunction operator could be interpreted in two ways. First, two actions are options in one command, as paraphrased in (19a). This is not a case where speech acts are disjoined. Second, two commands are really disjoined. The speaker gives one command or the other, and the speaker has not made up her mind, as Krifka claims. This reading is paraphrased as in (19b), but the sentence does not have this reading. This is pragmatically not allowed.8)

As for questions, there are disputes about whether they are disjoined. As discussed in (6), Belnap and Steel (1976) claim that questions are disjoined. However, it is not meant to ask one question or the other. The speaker asks whether the addressee has ever been to Sweden or to Germany. Krifka claims that this question can be answered by *yes* or *no*, but it is not the case. I suppose that the question has three answers: the addressee has been to Sweden, the addressee has been to Germany, the addressee has been to neither country.

However, there are clear cases where questions are not disjoined. Krifka (2001) and Szabolcsi (1997) discussed examples like the following:

(20) ??Which dish did Al make? Or, which dish did Bill make?

(20) is odd. The use of *or* has the effect of canceling the first question and the second question replaces it. In this respect, a forced interpretation is not a disjunction of two questions. The remaining issue is what difference there is between (6) and (20).

In Korean, no MPs are disjoined, just as no MPs are conjoined:9)

(21) a. *inho-ka ttena-ss-ta-kena mina-ka ttena-ss-ta.

8) This can be captured by the condition that an option mentioned in an imperative is a maximal option among those given in the context. Behind this condition lies the assumption that uttering an imperative is an act of decision, as claimed in Yeom (2018).

9) In Yeom (2019), I claim that this is related to the condition that an MP requires an act that exploits it. Matrix MPs must be associated with a speech act and there is only one speech act for more than one MP. Embedded MPs are associated with an act or situation expressed by an embedding verb. In (i) below, two embedded MPs are disjoined with -*kena*, and the disjunction operator is interpreted as a conjunction of two events of saying, one for each MP:

i. mino-nun inho-ka ttena-ss-ta-kena mina-ka ihonha-yess-ta-ko malha-yess-ta.
   Mino-top Inho-nom leave-pst-dec-or Mina-nom divorce-pst-dec-cmp say-pst-dec
   'Mino said whether Inho left or Mina got divorced.'

Inho-nom leave-pst-dec-or Mina-nom leave-pst-dec
'Inho left or Mina left.'

b. *inho-ka ttena-ss-nya-kena mina-ka ttena-ss-nya?
-int -int

c. *inho-ka ttena-la-kena mina-ka ttena-la!
-imp-or -imp

The disjunction operator *-kena* can conjoin embedded MPs, but not matrix MPs. In English, if MPs are disjoined, it is considered a case where speech acts are disjoined.

In Korean, *hokun* 'or' is the disjunction operator for speech acts. First I want to point out that statements are not disjoined due to the inferences from them:

(22) ??inho-ka ttena-ss-ta. hokun mina-ka ttena-ss-ta.
Inho-nom leave-pst-dec or Mina-nom leave-pst-dec
'Inho left. Or Mina left.'

By the first statement it is implicated that she believes that Inho left.[10] This is incompatible with the implicature from the second utterance that the speaker does not believe that Inho left (but that Inho might, and might not, have left). This leads to self-contradiction on the part of the speaker, which makes the discourse odd.

On the other hand, some questions are disjoined and some are not:

(23) *inho-ka eti iss-nya? hokun mina-nun eti iss-nya?
Inho-nom where be-int or Mina-top where be-int
'Where is Inho? Or, where is Mina?'

(24) inho-ka ttena-ss-nya? hokun mina-ka ttena-ss-nya?
Inho-nom leave-pst-int or Mina-nom leave-pst-int
'Did Inho leave? Or did Mina leave?'

10) Grice did not want to call the inference an implicature because it is not easily canceled:

i. ??John is happy. But I do not believe he is happy. (Moore's paradox)

This is called Moore's paradox. The term is attributed to Wittgenstein (1953). See also Moore (1993). We get the inference from the first statement and Grice's maxim of quality that the speaker believes that John is happy but it is incompatible with the second statement. If the inference were easily cancellable, the discourse would be fine. Still we can call the inference, or every inference from the Cooperative Principle by Grice, an implicature.

These observations show the same patterns observed in English. The question is why constituent questions and *Yes-No* questions behave differently in disjunction.

To get an idea of what is going on, consider the following example:

(25) ??inho-ka ttena-ss-nya? hokun mina-ka hayngpokha-nya?
Inho-nom leave-pst-int or Mina-nom be.happy-int
'Did Inho leave? Or is Mina happy?'

This is odd unless we assume special cases where the two questions are considered alternatives to each other. The disjoined questions in (24) can be considered alternatives to each other. For this reason, they can be merged into one question:[11)]

(26) {Inho left, Inho did not leave} OR {Mina left, Mina did not leave}
⇒ {Inho left, Mina left, (Neither left)}

It is assumed that each question is interpreted as a set of propositions corresponding to a set of possible answers, following Hamblin (1973), Karttunen (1977), etc. We can assume that the disjunction operator is interpreted to combine the two sets of possible answers to the union of them. Consequently, the context is structured into three partitions, under the assumption that each answer is exhaustive, just as Groenendijk and Stokhof (1984) do. In the resulting partition structure, the possible answers "Inho left" and "Mina left" are maintained. In this context, the addressee cannot answer "yes" or "no", as Krifka claims. The partition structure is more like a partition structure for a constituent question.

In this context, the issue is whether each of the original questions forms a separate speech act. It depends on the definition of a speech act. In the disjunction of the two questions, the first one is not a complete speech act, and the speaker completes her act of questioning by adding the second one. In this respect, each one is an incomplete speech act, and they are disjoined to form a complete speech act. An additional effect is that the speaker assumes that one of the two left, which possibly excludes the partition for "Neither left".

As shown in Yeom (2019), disjunction of questions is possible only if two questions are alternatives to each other. Otherwise, they are not interpreted as a single question, as in (25):

(27) {Inho left, Inho did not leave} OR {Mina is happy, Mina is not happy}

11) In (25), *hokun* can be replaced with *ani-myen* 'not.be-if'(= if not). This means the second question applies if the answer to the first question is negative. This leads to three partitions.

⇒ {Inho left, Mina is happy, Inho did not leave & Mina is not happy}

The partitions for *Inho left* and for *Mina is happy* cannot form a set of possible answers for one question, because they are not plausible alternatives to each other.

Disjoining two constituent questions in (23) does not form a single question:

(28) {Inho is in Seoul, Inho is in Busan, Inho is in Beijing, ⋯} OR
{Mina is in Seoul, Mina is in Busan, Mina is in Beijing, ⋯}
⇒ {Inho is in Seoul, Inho is in Busan, Inho is in Beijing, ⋯,
Mina is in Seoul, Mina is in Busan, Mina is in Beijing, ⋯}

The partitions for 'Inho is in Seoul' and for 'Mina is in Seoul' are not likely to form a single partition structure, because if they did, the question would have been *Who is in Seoul?* That is, the context is not structured to capture the meaning of the intended question.

From these discussions, we can make the following conclusion:

(29) Questions can be disjoined if the disjunction is re-interpreted as a single question.

Questions can be disjoined structurally, but the union of sets of alternative answers must form a set of alternative answers for one question. Due to this restriction, constituent questions are not disjoined. The reason is clear. A speech act is an action and the action is made only when the speaker has made a decision to do it. If we cannot get the meaning of one question, the speaker has not decided what to ask and it is odd to ask a question in such a context. This situation arises when a disjunction operator or a disjoined phrase has wide scope over a *wh*-phrase, which gets the scope of the interrogative Mood.

Imperatives are also disjoined, which is also discussed in Yeom (2019):

(30) A: icey naka-to.toy-e-yo?
now go.out-may-int-hon
'May I go out now?'
B: ani, emma-lul tow-ayahay. tongsayng-ul po-ala!
no mother-acc help-must brother-acc see-imp
hokun/ani-myen chengso-lul ha-yela!
or/not-if sweeping-acc do-imp
'No, you must help Mom. Take care of your brother! Or clean the rooms!'

In this context, A's utterance is included to make sure B's utterances are interpreted as commands. B rejects the permission for A to go out, and B's first imperative is interpreted as a command. But when the next imperative is added, the first one is not canceled but re-interpreted as an option in a longer command 'Take care of your brother or clean the rooms!'. Each of the two imperatives is an incomplete speech act that contributes to the formation of a complete command.[12)]

How can imperatives be disjoined? An imperative can have various meanings like a command, permission, request, warning, plea, a piece of advice, etc.,, as observed by Schmerling (1982), and the meaning of an imperative is contextually determined by selecting its modal base and quantificational force.[13)] This means that the meaning of modality in an imperative is flexible. If two imperatives in command readings are disjoined with *hokun*, they are interpreted as options of obligatory actions in one speech act of command.[14)]

(31) Two commands can be disjoined if they can be re-interpreted as one command.

From the discussions so far, the results can be summarized as follows:

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12) About this result, Krifka (2001) claims that the disjunction operator actually combines two propositions, not two commands, and that it becomes one command, as in (19a). However, there is a problem with Krifka's claim. Even if a disjunction of two speech acts can be re-interpreted as the meaning of one speech act, we cannot say that each of the two speech acts is a proposition. That is, the disjunction operator does not apply to propositions. A disjunction of two imperative MPs is (structurally) not the same as one imperative MP with a disjunction of two smaller phrases in it.

13) An imperative is assumed to denote a proposition with the meaning of modality. One typical property is that imperatives are used with various meanings: command, permission, warning, plea, wish, advice, invitation, concession, etc., as shown in (i), by selecting the ordering source and the quantificational force. See Schmerling (1982), Davies (1986), Palmer (1986), Bybee et al. (1994), and Xrakovskij (2001), Schwager (2006), Kaufmann (2012), etc.

i. a. Stop arguing!	(Command)
b. A(my): May I come in?	
B(ob): Sure, go ahead. Come this way!	(Permission)
c. A: Excuse me! How can I get to the station?	
B: Take a number 10 bus!	(Advice)
d. Be happy!	(Wish)
e. Please, don't rain tomorrow!	(Wish with no addressee)
f. Please pass me the butter!	(Request)
g. Okay, then sue me, if you have to!	(Concession)

14) When imperatives are disjoined, each imperative should be related to an alternative course of events for a given topic. On the other hand, the alternatives are given only implicitly in the context, while alternativeness in using a question is related to possible answers. For this reason, alternativeness is not captured in imperatives as clearly as in questions.

(32) a. $^{*}\upsilon(\varphi.) \vee \upsilon(\psi.) \neq \upsilon((\varphi \vee \psi).)$
b. $\upsilon(\varphi?) \vee \upsilon(\psi?) = \upsilon((\varphi \vee \psi)?)$ if ϕ and ψ are alternatives of each other.
c. $\upsilon(\varphi!) \vee \upsilon(\psi!) = \upsilon((\varphi \vee \psi)!)$ if ϕ and ψ are alternatives of each other.

Statements are not disjoined because they generate implicatures that are incompatible with each other. Questions and commands can be disjoined structurally if the disjuncts are alternatives to each other and they form a single complete speech act.

I discussed only cases where speech acts are disjoined structurally, and concluded that it is not a matter of structure whether questions and commands are disjoined. This implies that the restriction in (32) does not have to be based on the scope of the disjunction operator in the surface structure. Regardless of what phrases are disjoined, a disjunction operator is not allowed to have wide scope over a *wh*-phrase:

(33) a. ??inho-na mina-nun eti ka-ss-nya?
Inho-or Mina-top where go-pst-int
'Where did Inho or Mina go?'
b. ?*inho-nun eti ka-kena mina-nun eti ka-ss-nya?
Inho-top where go-or Mina-top where go-pst-int
'Where did Inho go or where did Mina go?'

(33a) can be interpreted as (33b), and both are odd. Structurally, the disjunction operator is realized within the Mood, but if it has wide scope over a *wh*-phrase, it has the same effect that the disjunction operator applies to the whole question and the sentence becomes odd.

If there is no *wh*-phrase, the disjunction operator is not forced to have wide scope over the Mood:

(34) a. inho-na mina-ka seoul-ey ka-ss-nya?
Inho-or Mina-nom Seoul-to go-pst-int
'Did Inho or Mina go to Seoul?'
b. inho-ka seoul-ey ka-kena mina-ka seoul-ey ka-ss-nya?
Inho-nom Seoul-to go-or Mina-nom Seoul-to go-pst-int
'Where did Inho go and where did Mina go?'

In this case, the sentence can be interpreted as a single question.

A conjunction or disjunction of two phrases can be interpreted with various scopes, regardless of what phrases are conjoined/disjoined. The restriction that a disjunction operator not have scope over the Mood is a pragmatic one, not a syntactic or semantic one. This pragmatic restriction should not impose any structural condition of what phrases are disjoined.

3. Quantifiers in interrogatives

In Section 2, I showed a conjunction/disjunction of phrases can have various scopes, even with the scope over the Mood if the meaning is pragmatically acceptable. These observations imply that in interpreting a sentence with a quantifier, disjunctive or not, the structural restriction on disjunction of speech acts should not be a factor.

Still scope interaction affects the acceptability of a quantifier in questions. Consider the following sentences where a *wh*-phrase has scope over a quantifier:

(35) nwu-ka {taypqwupwunuy, ceketo twu, kyewu twu} salam-ul manna-ss-nya?
who-nom most, at.least two, merely two people-acc meet-pst-int
'Who met {most, at least two, merely two} people?'

These are acceptable. The quantifiers could be scrambled over the *wh*-phrase so that they could have wide scope over it, but they are interpreted only with narrower scope:

(36) {taypqwupwunuy, ceketo twu, kyewu twu} salam-ul nwu-ka manna-ss-nya?
most, at.least two, merely two people-acc who-nom meet-pst-int
'Who met {most, at least two, merely two} people?'

The reason is that scrambled quantifiers can be interpreted in their original positions.[15] If quantifiers have scope over a *wh*-phrase with no scrambling involved, sentences are slightly odd:

(37) ?{taypwupwunuy, ceketo twu, kyewu twu} salam-i eti ka-ss-nya?
most, at.least two, merely two people-nom where go-pst-int
'Where did {most, at least two, merely two} people go?'

15) See Ahn (1990), Suh (1990), S. Kim (1991), Sohn (1995: 145), Son (2001). See also Huang (1982) for Chinese, and Hoji (1985) for Japanese.

From the surface scope relations, the quantifiers are expected to have wide scope over the *wh*-phrases, but we are forced to have the reversed scope relations different from the surface scope relations, which makes the sentences slightly odd. (37) can be compared with the following:

(38) eti-ey {taypwupwunuy, ceketo twu, kyewu twu} salam-i ka-ss-nya?
where-to most at.least two merely two people-nom go-pst-int
'Where did {most, at least two, merely two} people go?'

The *wh*-phrase is scrambled so that it can have wide scope over the quantifiers, and the sentences get better. From these observations, we can say that a quantifier cannot have scope over a *wh*-phrase.

If there is no *wh*-phrase in an interrogative, the quantifiers in (37) can be interpreted with narrower scope than the Mood and the sentence is interpreted as a single question:

(39) a. taypwupwunuy salam-i seoul-ey ka-ss-nya?
most people-nom Seoul-to go-pst-int
'Did most people go to Seoul?'
b. ceketo twu salam-i seoul-ey ka-ss-nya?
at.least two
c. kyewu twu salam-i seoul-ey ka-ss-nya?
merely two

Then why is a quantifier not allowed to have scope over a *wh*-phrase? This is related to the question of how the scope of a *wh*-phrase is determined. Its scope is not determined by the syntactic movement in Korean. In Korean, a *wh*-phrase in an interrogative is not subject to an island constraint:

(40) eti-ey sa(l)-nun salam-tul-i kacang hayngpokha-nya?
where-at live-adn people-pl-nom most be.happy-int
'Where$_i$ are people who live t_i are most happy?'

The NP *eti* 'where' came out of a syntactic island, and the sentence is fine. This implies a *wh*-phrase does not move. Still its scope is determined by the interrogative Mood. It is a separate issue how we get this result without assuming a syntactic movement. Somehow a *wh*-phrase is associated with the interrogative Mood.[16] I do not suppose that a quantifier is raised to a position over the interrogative Mood. If a quantifier is interpreted with wide scope over a *wh*-phrase, it must have scope over the interrogative Mood. Quantifiers simply cannot have scope over a Mood. Hence no reading of multiple speech acts (or, pair-list readings).

This is a little different from Krifka's (2001) claim that a question with a quantifier can have a pair-list reading only if the meaning can be defined as a conjunction of questions. He assumes that a quantifier can have scope over the interrogative Mood. However, we saw differences between a universal quantifier and a conjoined NP or a definite NP, as mentioned in (8), (9) and (10). See also fn. 3. The crucial observation is that in Korean even a universal quantifier does not allow a pair-list reading.

Next I am going to discuss imperatives, where no partition structure is relevant. An imperative is interpreted differently, and it is expected that a different restriction applies to conjunction or disjunction of phrases in imperatives and a different restriction on quantifiers.

4. Quantifiers in imperatives

4.1. Scope interactions in imperatives

We already saw that a different restriction works on quantifiers in imperatives. In imperatives, there is no scope asymmetry between subject positions and object positions. In (11), the quantifiers occur in the subject positions. Quantifiers in object positions show the same acceptability patterns:

(41) {??taypwupwun, ceketo twu kay, ??kyewu twu kay}-uy, sakwa-lul mek-ela!
most at.least two unit merely two unit}-of apple-acc eat-imp
'Eat {most, at least two, merely two} apples!'

In the object positions, *taypwupwunuy* 'most' and *kyewu* 'merely' make imperatives odd, and *ceketo* 'at least' does not.

Now I will show that a quantifier in an imperative also does not have wide scope over the Mood. To discuss scope interactions of quantifiers in imperatives, we need to discuss what semantic operators are involved in imperatives. In an imperative, there is a modal auxiliary. In Korean, it is not easy to find structural evidence for this assumption, but the meaning of

16) We can assume that a wh-phrase is focused and the interrogative Mood is a focus-sensitive operator, just as von Stechow (1990) assumes.

an imperative is not factual.[17] If an imperative has the meaning of a command or permission, the content is about a state of affairs in the future. If an imperative has the meaning of wish, the content is about bouletic alternatives to the actual world. Therefore, an imperative necessarily involves the meaning of modality.

Next we need to see if there is a scope interaction between a quantifier and modality in imperatives. It seems that a modal operator always has wide scope over a subject quantifier:

(42) twu salam-man ka-la!
two person-only go-imp
'Only two people go!' (It should not be the case that more people go.)

(43) a. ??The other people do **not have to** go, (= not [must [the other people go]])
b. The other people **must not** go. (= must [not [the other people go]])

(42) has the meaning of (43b), but not (43a). This means that the subject quantifier has narrower scope than the modal operator. A quantifier in the object position is also expected to have narrower scope than the modality operator.

On the other hand, a subject quantifier may, or may not, have wide scope over negation. To see this, I need to mention that *ma(l)* is used in imperatives, exhortatives, or deontic modality sentences with a narrower scope than the modality:[18]

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17) In English, we can find indirect structural evidence for modality. In an imperative, the first verb is always the base form, and as Potsdam (1998) showed, a *Do*-support is observed even if the first verb is *be* or an auxiliary verb *have*:

i. Please be thinking about me!
ii. a. Don't be afraid!
b. Please do have made that call by six o'clock!

These observations can be explained if a null modal operator is assumed in the T position:

iii. [$_{TP}$ you [$_T$ $\varnothing$ mod] not be afraid]
$\Rightarrow$ [$_{TP}$ you [$_T$ Do+$\varnothing$ mod +n't$_i$] t_i be afraid]

A null Mod is a bound morpheme, which requires a *Do*-Support. Without such a morpheme, *be/have* would move to the T position and a negative imperative would be as follows:

iv. *Be not afraid!

We can conclude that in English, there is an implicit bound morpheme of modality.

18) There are two forms of negation: a long-form and short-form negation. In an imperative, a short-form negation, which is of the form 'not V', is not allowed. In a long-form negation, the nominalizer *-ci* is

(44) ka-ci mal-{ala!, ca., aya-ha-n-ta.}
go-nml not-{imp, hor, must-do-impf-dec}
{Don't go, Let's not go, You must not go}

Here *-aya-ha* 'must-do' is a morpheme for modality. The negation morpheme *mal* 'not' occurs between the verb and the modal morpheme. A quantifier in an imperative shows scope interactions with negation:

(45) twu salam-man ka-ci ma((l-a)la)!
two people-only go-nml not-imp
a. More people go! (= must [not [only two people go]])
b. The other people except two people go! (= must [only two people [not go]])

(46) motwu ka-ci ma(l-ela)!
everybody go-nml not-imp
a. must [everybody [not go]]
b. must [not [everybody go]]

The quantifiers in (45) and (46) can have wide scope over, or narrower scope than, negation, but have narrower scope than the modal operator. To summarize the discussions, we can get the following scope relations:

(47) a. quantifier < neg < modality
b. neg < quantifier < modality

Assuming that a Mood has wide scope over modality, no quantifier can have wide scope over the Mood. This indicates that the restrictions on quantifiers in imperatives cannot be explained based on conjunction or disjunction of speech acts.[19)]

4.2. Meaning of an imperative and quantifiers

The question is what factor(s) of an imperative restrict(s) a quantifier in it. To get the answer, we need to know how an imperative is interpreted. The question of what an

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used, and it gets the form "VP-*ci* {*anh*, *mos*, *ma(l)*}".

19) In this respect, Portner's (2007) claim that the meaning of an imperative is a member of a *To-do*-List as denoting a property of individuals. The meaning of an imperative includes more than just a property of individuals.

imperative means is more fundamental than the topic of this paper, and beyond the scope of this paper. See Yeom (2018) for more extensive discussions of the semantics of imperatives. Here I will just sketch the idea only to the extent that I can explain the restrictions on quantifiers in imperatives.

One main characteristic of imperatives is that it changes the actual world:

(48) A: I know I have to stay here until 5, but my mother came to see me.
B: Okay, then go home now!

A's statement presupposes that A must stay here till 5, and B admits it. But B has changed his mind and makes an utterance that overrides the presupposition. I will call this a **presupposition override**.[20] A presupposition override arises due to the speaker's decision (and the addressee's acceptance of the decision). See Yeom (2018) for more discussions of presupposition override. The content of the decision is the meaning of the sentence including modality. In (48), the content of the imperative is 'A may go home now'. The content itself can be said to be descriptive, but the imperative Mood makes the sentence non-descriptive, that is, non-propositional. Therefore the imperative in (48) changes a possible world satisfying a presupposition to a possible world not satisfying it.

Reflecting this idea, the meaning of an imperative can be defined as follows:

(49) $[\![\phi!]\!]^c = \{\langle w, w'\rangle \mid w \in cg,\ w' \in max_{cg,\ \varphi'}(\{w'' \in W \mid w'' \simeq_{ut} w\})\}$
$c = \langle sp(eaker),\ ad(dressee),\ u(tterance)t(ime),\ c(ommon)g(round)\rangle$
$w'' \simeq_{ut} w$: w'' has the same history as w until ut
$max_{cg,\ \varphi}(X) = \{w'' \in X$: for any $w''' \in X$,
$\{q \mid cg \subseteq q,\ w''' \in (\varphi' \cap q),\ \neg(\neg\varphi' \subseteq q)\} \subseteq \{q: cg \subseteq q,\ w'' \in (\varphi' \cap q),\ \neg(\neg\varphi' \subseteq q)\}\}$

An imperative changes a world w to w′ such that w′ is a ϕ-world with the same history as w till the utterance time and w′ is maximally similar to w with respect to the information in the common ground. This means that an imperative changes the context, if the addressee

20) This needs to be distinguished from presupposition cancellation, which occurs when an operator that can cancel a presupposition:

i. a. John's father is not coming, because his father is dead.
b. It is possible that John has stopped smoking, but it is possible that he has never smoked.

In (ia), the presupposition that John has a father is canceled due to the negation operator. In (ib), it is the possibility operator that suspends the presupposition that John has been smoking.

accepts the decision, to a context that supports it.[21)]

The semantics of an imperative requires a different way of interpreting a quantifier. In general a quantifier is interpreted in a propositional sentence as follows:

(50) a. twu salam-i ttena-ss-ta.
two person-nom leave-pst-dec
b. $\{w \in W \mid \#(\{x \mid person^{w}(x) \wedge left^{w}(x)\}) = 2\}$
c. ttena-n salam-uy swu-ka twul-i-ta.
leave-adn person-of number-nom two-be-dec
'The number of people who left was 2.'
(51) a. twu salam-i ttena-ss-nya?
two person-nom leave-pst-int
b. $\{\{w \in W \mid \#(\{x \mid person^{w}(x) \wedge left^{w}(x)\}) = 2\},$
$\{w \in W \mid \#(\{x \mid person^{w}(x) \wedge left^{w}(x)\}) \neq 2\}\}$[22)]
c. ttena-n salam-uyswu-ka twul-i-nya?
-int (Cf. 50c)

(50a) is interpreted as (50b), namely that the number of people who left was two. This can be expressed as (50c). The question in (51) can be interpreted in a similar way, and the same meaning can be expressed as in (51c). This can be intuitively correct. However, this way of interpretation is not acceptable in imperatives. Interpreting (52a) as (52b) is not adequate:

(52) a. twu salam-i ttena-la!
two person-nom leave-imp
'Two of you leave!'
b. ??ttena-l salam-uy swu-ka twul-i-ela!
leave-adn people-of number-nom two-be-imp
'The number of people who are leaving be 2!'

An imperative has the meaning of a world change by the speaker's decision and the

21) There are cases where an imperative does not change the world, and in such cases, it is also the speaker's decision not to change the world:

i. A: I will stay.
B: Okay, you stay!

22) Here I assume that the numeral has the 'exactly' reading. If it is assumed to have the 'at least' reading, the partitions can be different. See Breheny (2008), and Yeom (2017) for pragmatic analyses of numerals.

addressee's accepting the decision. This means that the content should be able to become true by them.[23] The number of people who left is not a matter that can be true simply by the speaker's decision. This restriction is generally expressed as an aspectual restriction on the predicate used in an imperative.

This leads to the issue of how a quantifier is interpreted in an imperative. In an imperative, a quantifier with *twu* 'two' needs to be interpreted in a different way.[24] A more plausible interpretation of (52a) is as follows, assuming that the addressees are a, b, and c, which is represented as a sum individual "a&b&c":

(53) (a&b leave ∨ b&c leave ∨ a&c leave)!

Here "a&b leave" can be imposed as an action on the sum individual a&b. It can be considered a member of a&b's *To-do*-List in the sense of Portner's (2005, 2007) analysis. We can say similar things about "b&c leave" and "a&c leave". The three cases are disjoined.

More generally, the meaning of (53) can be represented as follows:

(54) $[\![$twu salam-i ttena-la!$]\!]$ = $(\bigvee\{leave(x) \mid x \sqsubseteq c_A \wedge \#(x)=2\})!$
($\bigvee\{p, q, \cdots\} =_{def} (p \vee q \vee \cdots)$; x ranges over sum individuals;
#(x): the number of atoms in x, or the amount of a material x by a measure unit)

"$\bigvee\{p, q, ...\}$" is equivalent to "$(p \vee q \vee \cdots)$". The variable x ranges over sum individuals. Other quantifiers are interpreted as follows:

(55) $[\![all]\!] = \lambda P \lambda Q[\bigvee\{Q(x) \mid y'=max(\{y \mid P(y)\}), x \sqsubseteq y', y' \sqsubseteq x\}]$
$[\![many]\!] = \lambda P \lambda Q[\bigvee\{Q(x) \mid y'=max(\{y \mid P(y)\}), x \sqsubseteq y', \#(x)>n, n \text{ is a large number}\}]$
$[\![some]\!] = \lambda P \lambda Q[\bigvee\{Q(x) \mid y'=max(\{y \mid P(y)\}), x \sqsubseteq y', \#(x)>0\}]$

'$max(\{x, y, \cdots\})$' is a maximal element in the set $\{x, y, \cdots\}$. The meanings of quantifiers are defined as a disjunction of the meanings of the nuclear scope, each of which corresponds to a member of a set of sum individuals determined by the restrictor. If the quantifier is a

23) For this reason, Portner (2005, 2007) claims that the meaning of an imperative is a member of a *To-do*-List for the addressee. He is trying to capture that an imperative involves an action on the part of the addressee, though I do not agree with his analysis of imperatives.

24) One exceptional case is one where an imperative is used with the meaning of a wish:

i. Please be pretty! (in a context where the speaker is going to a blind date) universal quantifier, there is only one member in the set of meanings of the nuclear scope. We can assume that $\bigvee\{p\} = p$, where p is a proposition. Non-universal quantifiers generally generate a non-singular set of meanings of the nuclear scope, and the disjunction of them is well-defined.

4.3. Restrictions of *kyewu*-quantifiers

An imperative is non-propositional and can change the actual world by the speaker's decision (and the addressee's acceptance). This imposes a restriction on the content of the meaning of an imperative. We saw in (11) that in imperatives, a quantifier with *kyewu* 'merely' is not acceptable. That example is repeated here:

(56) ??kyewu twu salam-i ka-la!
merely two person-nom go-imp
'Merely two people go!'

We can explain this observation by resorting to the meanings of an imperative and the expression *kyewu*. *Kyewu* roughly has the meaning of 'smaller than expected', which I take to be a conventional implicature (CI).[25] The meaning of an NP with the expression is as follows:

(57) $[\![\text{kyewu } n\ \alpha]\!] = \lambda P[\bigvee\{P(x) \mid y' \in \max\{y \mid [\![\alpha]\!](y)\},\ x \sqsubseteq y',\ \#(x)=n\} \wedge$
$\ll$n is smaller than expected $\gg$]
$\ll\ \gg$: Conventional Implicature

The CI from *kyewu* combines with the semantic meaning to make a strong meaning. In (12), for example, the sentence with *kyewu* has the following meaning:

(58) [[kyewu twu salam-i ttena-ss-ta]] =
$\lambda w[\bigvee\{\text{left}^w(x) \mid y' \in \max\{y \mid \text{people}^w(y)\},\ x \sqsubseteq y',\ \#(x)=2\} \wedge$
$\ll$2 is smaller than expected in w $\gg$]

25) In this discussion, I follow Grice's notion of conventional implicature (CI). As one anonymous reviewer pointed out, Potts (2005) claims that CIs are scopeless, but his notion of CI is different from Grice's in that he assumes CIs arise from speaker-oriented expressions. Moreover, as pointed out in Amaral, Roberts and Smith (2007), "... in the cases where CIs are anchored to an agent other than the speaker, they do appear to take narrow scope relative to the embedding attitude predicate." (Amaral, Roberts and Smith (2007), p. 25)

The sentence is propositional and it is fine to conjoin its semantic meaning and its CI. The CI tends not to be projected:[26)]

(59) inho-nun kyewu twu salam-i ttena-ss-ta-ko sayngkakha-n-ta.
Inho-top merely two people-nom leave-pst-dec-cmp think-impf-dec
'Inho thinks that merely two people left.'
a. ??The number of people who left was smaller than expected.
b. Inho thinks that the number of people who left was smaller than expected.

The CI belongs to the propositional attitude context.

This implies that in an imperative, the CI has narrower scope than the modal operator. That is, it should be part of the command or permission. Therefore, (56) roughly has the meaning of (60):

(60) ??twu salam-i ttena-ko 2-ka kitay-pota cek-un swu-i-ela!
two person-nom leave-and 2-nom expect-than little-adn number-be-imp
'Two people go and 2 be a smaller number than expected!'

This sentence is odd because the meaning that the number of the people who are leaving needs to be smaller than expected cannot be the content of a command. This meaning is formally represented as follows:

(61) $[\![(56)]\!] = (\bigvee\{\text{leave}(x) \mid x \sqsubseteq c_A,\ \#(x)=2\} \wedge \ll 2 \text{ is smaller than expected} \gg)!$

I assume that the subject of an imperative is normally a subset of the addressee(s).[27)] In the meaning representation in (61), the implicature should be part of the command, but it cannot be true simply by the speaker's decision in the addressee's action. This makes the imperative

26) This shows that CIs must be distinguished from presuppositions, which tend to be projected. In (i) below, the presupposition that arises in the embedded clause can be projected and we can assume that Inho has been smoking.

i. Mina thinks that Inho stopped smoking.
» Inho has been smoking.

27) This is not quite correct. Some imperatives have no subject, as in (ia), and some have a subject that is not part of the addressee, as in (ib).

i. a. Please, don't rain tomorrow!
b. You go for help and the baby stay with me!

odd.

If an imperative is interpreted as a wish, the content does not have to involve any action. Thus it might be claimed that the implicature can be interpreted as a wish and the rest can be interpreted as a command. However, it was shown by Portner (2005, 2007) that imperatives in a context are interpreted with the same type of modality.

(62) a. Be there at least two hours early.
b. Then, have a bite to eat.

If the first sentence is interpreted as a suggestion, the second is too. It is odd to interpret the first one as a command and the second as a suggestion. The two imperatives must be interpreted with the same type of modality. This means that in a command, the content for a wish cannot be included.[28)]

I tried to explain the restriction on *kyewu*-quantifiers based on CIs they generate. To show that this is the correct explanation, I can give other cases where CIs affect acceptability of imperatives:

(63) ?*hwanca-lul wuyhay kitoha-l.ppwun-i-ela!
patients-acc for pray-only-be-imp
'Only pray for the patients!'
(64) ??kikkeshayya sakwa twu kay-lul mek-ela!
at.most apple two piece-acc eat-imp
'Eat at most two apples!'

The morphemes *-l.ppwun* 'merely' and *kikkeshayya* 'at most' add the CI that the situation described by the rest of the sentence is not sufficient. The implicature does not involve the addressee's action, and it cannot be part of a command or permission.

A different example is the use of *-ciman* 'but':

(65) ??sey salam-kwa ssawu-**ciman** iki-ela!
3 people-with fight-but win-imp

28) If an imperative is interpreted as a wish, *kyewu* is acceptable:

i. ceypal kyewu twu mwuncey-man nam.aiss-ela!
please merely two problem-only remain-imp
'Please merely two problems remain!'

'Fight with three people but win!'

It adds the CI that if the addressee fights with three people, it is expected that she will lose and that the expectation is not satisfied. The meaning of concession cannot be part of the meaning of an imperative. This can be contrasted with the following:

(66) sey salam-kwa ssawu-**ko** iki-ela!
3 people-with fight-and win-imp
'Fight with three people and win!'

The connective *-ko* 'and' does not add such a CI, and the imperative is acceptable.

If an expression yields a CI that involves no action of the addressee, it is not used in an imperative. *Kyewu* is such an expression.

4.4. Restrictions on *taypwupwunuy*-quantifiers

A use of *taypwupwunuy* 'most' makes an imperative odd. This cannot be explained based on a CI. What meaning prohibits the quantifier from being used in an imperative? To get an answer, I will consider frequency adverbs.

There are frequency adverbs that correspond to adnominal quantificational expressions:

(67)

motun	'all/every'	hangsang	'always'
taypwupwunuy	'most'	taykay	'usually'
manhun	'many'	cacwu	'frequently'
yakkanuy	'a little'	kakkum	'sometimes'
twu	'two'	twu pen	'twice'

The adverbs show the same restrictions as the adnominal quantificational expressions in imperatives. The sentences with *taykay* 'usually' and *kyewu* 'merely' are unacceptable. I will focus on *taykay*.

(68) {hangsang, ??taykay, cacwu, kakkum, (??kyewu) twu pen} sakwa-lul mek-ela!
{always, usually, often, sometimes, (merely) two times} apple-acc eat-imp
'Eat apples {always, usually, often sometimes, twice}!

The fact that frequency adverbs show the same acceptability shows that the relevant factor is not syntactic nor peculiar to a particular morpheme. A different morpheme with a similar meaning shows the same restriction:

(69) ??{pothong, ilpancekulo, cwulo} sakwa-lul mek-ela!
{usually, generally, mainly} apple-acc eat-imp
'{Usually, Generally, Mainly} eat an apple!'

We have to explain why some frequency adverbs like *cacwu* 'often' and *kakkum* 'sometimes' can be used in imperatives. One of the properties of these adverbs is that the situations relevant are distributed roughly evenly. But the adverbs that are not allowed in imperatives do not require temporally even distributions:

(70) a. ku senswu-nun cenpanki-ey-nun antha-lul mos chi-taka hwupanki-ey-man
the player-top first.half-at-top safe.hit-acc not hit-and second.half-at-only
chi-ess-ta.
hit-pst-dec
'The player did not make hits in the first half of the season and made hits only in the second half.'
b. cencheycekulo {*cacwu*, *kakkum*} antha-lul chi-n-ta.
overall {often, sometimes} safe.hit-acc hit-impf-dec
'Overall he makes hits {often, sometimes}.'
c. cencheycekulo coh-un kong-i o-myen *taykay* antha-lul chi-n-ta.
overall good-adn ball-nom come-if usually safe.hit-acc hit-impf-dec
'Overall, if a good ball comes, he generally makes a hit.'

In (70a), it is specified that the player did not make temporally evenly distributed hits. It can be followed by (70c), but not by (70b). This indicates that the frequency adverbs *cacwu* 'often' and *kakkum* 'sometimes' require temporally evenly distributed hits, but the frequency adverb *taykay* does not.

A sentence with *cacwu* and *kakkum* can apply to a limited time, but a sentence with *taykay* cannot:

(71) han sikan-tongan-un {cacwu, kakkum, ??taykay} nwun-ul kkampaki-ess-ta.
one hour-during-top often sometimes generally eye-acc blink-pst-dec
'During one hour, she blinked her eyes {often, sometimes, generally).

This indicates that a sentence with the adverb *taykay* 'generally' expresses genericity. This implies that a sentence with the adverb applies to an unlimited period of time and describes a *general* situation from the perspective of a long-term period.

Now consider the meaning of an imperative. An imperative can be used in a episodic or generic context:

(72) A: I have a date tomorrow. What shall I do? (episodic)
B: Ask her what she wants to do (tomorrow)!

(73) A: I don't know what to do when I have a date with a girl. (generic)
B: Ask your date what she wants to do (whenever there is a date with a girl)!

(72) is an imperative involving one-time action, while (73) is an imperative applying to a general situation type. This does not mean an imperative can be a generic sentence. If an overt generic operator is inserted, the imperative becomes odd:

(74) ??{Generally, Usually} ask your date what she wants to do!

This implies that even in a generic imperative like the case in (73), the action expressed by the imperative is a particular one in a local context where a certain condition is met. In (73), the condition is that A had a date with a girl. In (73), the addressee must perform a concrete action of asking his date whenever he has a date with a girl. He does not perform a generic action of asking his date. In fact, there is nothing like a generic action. The genericity is provided as a context in which the imperative applies. This can be represented as follows:

(75) $$\underbrace{\mathrm{Gen}(\lambda s[\alpha(s)])(\underbrace{\lambda s'[\beta(s)]}_{\checkmark \mathrm{imp}})}_{^{*}\mathrm{imp}}$$

An imperative itself cannot become a generic sentence, but it can occur in the position of a nuclear scope of a generic operator. The imperative in (73) is in such a use. This allows us to explain why an imperative with *taykay* is odd. *Taykay* 'generally' is a generic operator. It does not occur in the nuclear scope of a generic operator.

The adnominal quantity expression *taypwupwunuy* 'most' corresponds to the frequency adverb *taykay* 'generally'. It is generally interpreted as follows:

(76) $[\![\text{most}]\!] = \lambda P\lambda Q[\bigvee\{Q(x) \mid y'=\max(\{y \mid P(y)\}),\ x\sqsubseteq y',\ \#(x)/\#(y') > ½\}]$

That is, it is assumed to be equivalent to 'more than half'. If we accept this assumption, there is no reason that the quantifier is not permitted in an imperative:

(77) a. pan isang-uy sakwa-lul mek-ela!
half more-of apple-acc eat-imp
'Eat more than half the apples!'
b. pan isang-uy salam-i ttena-la!
half more-of people-nom leave-imp
'More than half of you leave!'

This shows that *taypwupwunuy* 'most' is different from *pan isang* 'more than half'.

Considering the corresponding adverb *taykay*, we can suppose that *taypwupwunuy* has the meaning of genericity. One problem with this idea is that it can be used in a sentence that describes an episodic situation, as we saw in (12). The only way to resolve these conflicting requirements is to assume that it conveys both an episodic and a generic meaning together. Then the semantics is compatible with an episodic situation but it also has the meaning of genericity.

To represent the meaning of *taypwupwunuy*, I need to explain how the meaning of genericity is expressed. The exact meaning of genericity is a little complex, but I will simply assume that it is universal quantification over normal worlds. See Krifka et al. (1995) as well as Pellerier and Asher (1997) for further discussions of genericity.

(78) $[\![\text{taypwupwunuy } \alpha]\!]$
$= \lambda w\lambda Q[\text{(i) } \bigvee\{Q(x) \mid y'=\max(\{y \mid [\![\alpha]\!]^{w}(y)\}),\ x\sqsubseteq y',\ \#(x)/\#(y') > ½\} \wedge$
$\text{(ii) } \forall w''[w''\in NORM_{w}(\{w' \mid \max(\{y \mid [\![\alpha]\!]^{w'}(y)\}) \doteqdot \max(\{y \mid [\![\alpha]\!]^{w}(y)\})\})$
$\rightarrow [\bigvee\{Q(x) \mid x\sqsubseteq y'',\ \#(x)/\#(y'') > ½\}]]$
$NORM_{w}(p)$ = a set of normal worlds with respect to the true generic sentences in w that are compatible with p

In (78), the meaning in (i) is an episodic meaning of the adnominal quantifier, and the meaning in (ii) reflects the meaning of genericity. The meaning of genericity expresses that the sentence would also hold in all normal worlds.

With the meaning of (78), (12) with *taypwupwunuy* means that most people left and that it is normal. The meaning of (78) also allows us to explain why an imperative with *taypwupwunuy* is odd. The meaning of genericity is incompatible with the meaning of an imperative, as in (74).

5. Conclusion

In this paper, I showed that in Korean, interrogatives and imperatives can be disjoined structurally, but one condition is that a disjunction of interrogatives or imperatives be re-interpreted as one question or command. This implies that speech acts cannot be disjoined semantically and pragmatically.

On the other hand, this restriction on disjunction of speech acts is different from the condition on quantifiers in interrogatives/imperatives. In interrogatives, a quantifier cannot have scope over a *wh*-phrase because the scope of a *wh*-phrase is determined by the interrogative Mood but a quantifier cannot have scope over the Mood.

Restrictions on quantifiers in imperatives are different from restrictions on quantifiers in imperatives. An imperative is a world changer by the speaker's decision and the addressee's accepting the decision. Some quantifiers are not allowed in imperatives because the CIs they generate are incompatible with the semantics of imperatives. Some quantifiers have the meaning of genericity, and an imperative is not compatible with the meaning of genericity.

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